

HER2-targeted therapies in gastric cancer.

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Molecular targeted therapy of cancer has always been the focus of clinicians. Among those therapeutic targets, the human epidermal growth factor receptor-2 (HER-2) signaling pathway is one of the most popular targets for translational research in cancer. However, unlike prospect in breast cancer, HER-2 inhibitor trastuzumab is the only molecular targeted drug approved by US Food and Drug Administration (FDA) for the first-line treatment of HER-2 positive advanced gastric cancer. On this basis, a variety of novel HER2- targeted drugs for gastric cancer are under development, and related clinical researches are in full swing, including small molecular kinase inhibitors (e.g., afatinib, neratinib, pyrotinib), antibody-drug conjugates (e.g., DS-8201a, RC48-ADC) and other novel therapies (e.g., ZW25, CAR-T, BVAC-B). In this study, we will summarize the recent advances in anti-HER-2 agents, potential mechanisms of resistance to HER2-targeted therapy in HER2-positive gastric cancer. We will also discuss the future prospects of potential strategies to overcome anti-HER-2 resistance and development of novel anti-HER-2 approaches for the treatment of HER2-positive gastric cancer patients.

In utero treatment of congenital cytomegalovirus infection with valganciclovir: an observational study on safety and effectiveness.

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The treatment of congenital cytomegalovirus (CMV) infection is usually administered to neonates after birth; however, it can be anticipated during the prenatal period by treating pregnant women in order to reduce the severity of the congenital disease. The most commonly used treatment for CMV during pregnancy is valaciclovir; however, valganciclovir has a higher potency against CMV and is the first choice for neonates with congenital CMV disease. We investigated neonatal and maternal safety of tertiary prevention in infected fetuses showing ultrasound features of infection using valganciclovir. Retrospective cohort study of pregnant women and their symptomatic infected fetuses taking valganciclovir, 3 × 450 mg per day. All fetuses presented at least one prenatal feature on ultrasound. We assessed fetal/neonatal and maternal safety, as well as neonatal efficacy of treatment. The main outcome was neutropenia. Secondary outcomes included other haematological side effects, symptoms at birth and neonatal CMV-PCR was positive. Seventeen women with singleton pregnancies received valganciclovir from a median (IQR) of 27.1 (26.0-30.3) to 11.6 (6.5-12.9) weeks of gestation. No neonatal neutropenia was reported. One pregnancy was terminated for severe features. Three newborns (18%) were asymptomatic at birth, including one with negative CMV-PCR from blood and saliva. CMV-PCR was positive for 12/13 symptomatic newborns, with a median (IQR) log₁₀ viral load of 3.36 (3.30-4.20), 4.03 (1.75-4.27) and 3.04 (0.00-3.40) log₁₀ copies/mL in blood, urine and saliva, respectively. Tertiary prevention by valganciclovir appears to be well tolerated for both fetus and mother. However, more extensive trials accompanied by long-term follow-up are needed.

Prevention strategies for congenital cytomegalovirus infection.

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Cytomegalovirus (CMV) is the most common viral cause of congenital infection, occurring in

approximately 1-2% of live births worldwide. Given our increasing knowledge of risk, advances in the identification of maternal infection, and the extremely limited options for the treatment of fetal infection, the prevention is a promising direction for research efforts. Recently, there have been several exciting studies assessing different ways of preventing congenital infection in the fetus and one in particular has focused on the use of valaciclovir. A recent study reported a 71% reduction in vertical transmission of CMV with the use of oral valaciclovir following maternal primary CMV infection early in pregnancy. The clinical impact of this study could be enormous and it has particular implications for considerations around maternal serological screening in the first trimester of pregnancy. Further research assessing behaviour modifications during early pregnancy could also provide evidence for an effective primary prevention technique. Prevention of congenital CMV infection, whether primary, secondary or tertiary, is possible, however, there are barriers to its utilisation in a clinical setting. The main limitation is the requirement for early, effective and large-scale serological screening of mothers to detect asymptomatic primary infection.

Describing the Impact of Maternal Hyperimmune Globulin and Valacyclovir on the Outcomes of Cytomegalovirus Infection in Pregnancy: A Systematic Review.

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Cytomegalovirus (CMV) is the leading infectious cause of congenital neurological disabilities. Valacyclovir and CMV hyperimmune globulin (HIG) may reduce vertical transmission and sequelae in neonates. A systematic review on valacyclovir and CMV HIG in preventing vertical transmission or reducing sequelae in neonates was conducted to 3 September 2021. Valacyclovir as a preventive strategy was supported by a well-conducted randomized controlled trial. Evidence supporting valacyclovir as a treatment strategy was limited to observational studies at moderate risk of bias. CMV HIG was not supported as a preventive strategy in 2 randomized controlled trials, which contrasted with observational studies. Evidence favoring CMV HIG as a treatment strategy was limited to observational studies at moderate risk of bias. The role of valacyclovir and CMV HIG in CMV infection in pregnancy is still being defined. Valacyclovir to prevent vertical transmission has the highest quality evidence in favor of use.

Valacyclovir in primary maternal CMV infection for prevention of vertical transmission: A case-series.

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No treatment is currently approved for cytomegalovirus infection in pregnancy. Valacyclovir has been studied in symptomatic cytomegalovirus infected fetuses and seems to reduce the risk of serious sequelae. We used off-label valacyclovir on pregnant women with primary cytomegalovirus infection to reduce the risk of fetal infection. We treated 12 pregnant women with 8 g/day valacyclovir after diagnosis of cytomegalovirus infection until amniocentesis. We continued treatment until delivery in case of fetal infection. We periodically performed serology and virology tests on the women from referral until delivery and monitored them for adverse effects while on treatment. All women underwent late amniocentesis. We followed up infants for 5-28 months. At the time of amniocentesis, we observed a transmission rate of 17 %, and at birth we observed a transmission rate of 42 %. Two women with negative amniocentesis and infected newborns had viremia reactivation after valacyclovir discontinuation. We observed no

symptomatic infections at birth and one isolated sensory-neural hearing loss at follow-up. This is the first series of antiviral treatment in women with a diagnosis of cytomegalovirus infection before amniocentesis. Valacyclovir may control cytomegalovirus infection while it is administered and reduce transmission at amniocentesis. Late transmission after treatment discontinuation is a risk. We advocate the need for a controlled trial of valacyclovir therapy starting from diagnosis of maternal infection until delivery, regardless of prenatal diagnosis of infection.

Guinea Pig Cytomegalovirus (GPCMV): A Model for the Study of the Prevention and Treatment of Maternal-Fetal Transmission.

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A major public health challenge today is the problem of congenital cytomegalovirus (CMV) transmission. Maternal-fetal CMV infections are common, occurring in 0.5-2% of pregnancies, and these infections often lead to long-term injury of the newborn infant. In spite of the well-recognized burden that these infections place on society, there are as yet no clearly established interventions available to prevent transmission of CMV. In order to study potential interventions, such as vaccines or antiviral therapies, an animal model of congenital CMV transmission is required. The best small animal model of CMV transmission is the guinea pig cytomegalovirus (GPCMV) model. This article summarizes the GPCMV model, putting it into the larger context of how studies in this system have relevance to human health. An emphasis is placed on how the vertical transmission of GPCMV recapitulates the pathogenesis of congenital CMV in infants, making this a uniquely well-suited model for the study of potential CMV vaccines.

Maternal-fetal cytomegalovirus infection: from diagnosis to therapy.

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Cytomegalovirus (CMV) is the most common and serious congenital infection, because it occurs after both primary and recurrent infection in pregnancy and is a major cause of childhood deafness and neurological handicap. Fetal transmission generally occurs in 30-60% of women acquiring a primary infection or in 0.5-2% of women with preconceptional immunity. Following primary maternal infection, approximately one-third of the CMV-infected infants will have disease at birth or develop severe sequelae, which include convulsive or spastic syndromes, mental retardation and auditory and visual impairment. Routine antepartum and in pregnancy serological screening should be essential for avoiding doubtful interpretations of CMV-IgM results in pregnancy. Detection of quantitative CMV DNA in the amniotic fluid and accurate ultrasound examinations are needed in pregnant women with suspected primary infection. Being not yet available a vaccine and uncertain the results of postnatal ganciclovir therapy, CMV hyperimmunoglobulin appears to be the only safe and valid approach for prevention of congenital CMV disease.

A systematic review of family history, race/ethnicity, and genetic risk on prostate cancer detection and outcomes: Considerations in PSA-based screening.

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To investigate the role of family history, race/ethnicity, and genetics in prostate cancer (PCa) screening. We conducted a systematic review of articles from January 2013 through September 2023 that focused on the association of race/ethnicity and genetic factors on PCa detection. Of 10,815 studies, we identified 43 that fulfilled our pre-determined PICO (Patient, Intervention, Comparison and Outcome) criteria. Men with ≥ 1 first-degree relative(s) with PCa are at increased risk of PCa, even with negative imaging and/or benign prostate biopsy. Black men have higher PCa risk, while Asian men have lower risk. Most of the differences in risks are attributable to environmental and socioeconomic factors; however, genetic differences may play a role. Among numerous pathogenic variants that increase PCa risk, BRCA2, MSH2, and HOXB13 mutations confer the highest risk of PCa. Polygenic risk score (PRS) models identify men at higher PCa risk for a given age and PSA; these models improve when considering other clinical factors and when the model population matches the study population's ancestry. Family history of PCa, race/ethnicity, pathogenic variants (particularly BRCA2, MSH2, and HOXB13), and PRS are associated with increased PCa risk and should be considered in shared decision-making to determine PCa screening regimens.

Prostate cancer screening-when to start and how to screen?

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Prostate-specific antigen (PSA) screening reduces prostate cancer (PCa) mortality; however such screening may lead to harm in terms of overdiagnosis and overtreatment. Therefore, upfront shared decision making involving a discussion about pros and cons between a physician and a patient is crucial. Total PSA remains the most commonly used screening tool and is a strong predictor of future life-threatening PCa. Currently there is no strong consensus on the age at which to start PSA screening. Most guidelines recommend PSA screening to start no later than at age 55 and involve well-informed men in good health and a life expectancy of at least 10-15 years. Some suggest to start screening in early midlife for men with familial predisposition and men of African-American descent. Others suggest starting conversations at age 45 for all men. Re-screening intervals can be risk-stratified as guided by the man's age, general health and PSA-value; longer intervals for those at lower risk and shorter intervals for those at higher risk. Overdiagnosis and unnecessary biopsies can be reduced using reflex tests. Magnetic resonance imaging in the pre-diagnostic setting holds promise in pilot studies and large-scale prospective studies are ongoing.

PSA quo vadis? It is reasonable to start with prostate-specific antigen testing at the age of 40!

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It is common to start with PSA (prostate-specific antigen)-testing at the age of 50. If patients with a PSA value greater than 4 ng/mL should be considered for prostate biopsy, approximately 20% of all men undergoing test are considered for biopsy at the time of first early-detection examination. We have screened 2,592 asymptomatic men younger than 45 years. With a short questionnaire, we assessed age, body mass index (BMI), concomitant diseases, last sexual intercourse, and last bicycle riding. We compared this cohort with a group of 11,656 men aged 45-75 years from a nationwide prostate cancer

screening trial. In this cohort, only 4 men with a PSA value greater than 4 ng/mL and 10 with a PSA greater than 3 ng/mL were identified. More than 99% of all men younger than 45 years had a PSA lesser than 4 ng/mL. Sexual intercourse, bicycle riding, or BMI had a significant but minimal influence on the PSA value. It is reasonable to start with PSA testing at the age of 40 years. The advantage of screening younger patients is that almost no one should be considered for biopsy at the time of first early-detection examination. We identified a baseline value at which only a minimal influence was exerted by benign prostatic hypertrophy. In comparison with many current guidelines, we gained a lead time of 10 years for observation of PSA dynamics. The importance of PSA velocity for stratification of patients at risk for development of significant prostate cancer will grow.
